Iron Ore Purity Prediction

Tholang Chonelanga

06/01/2021

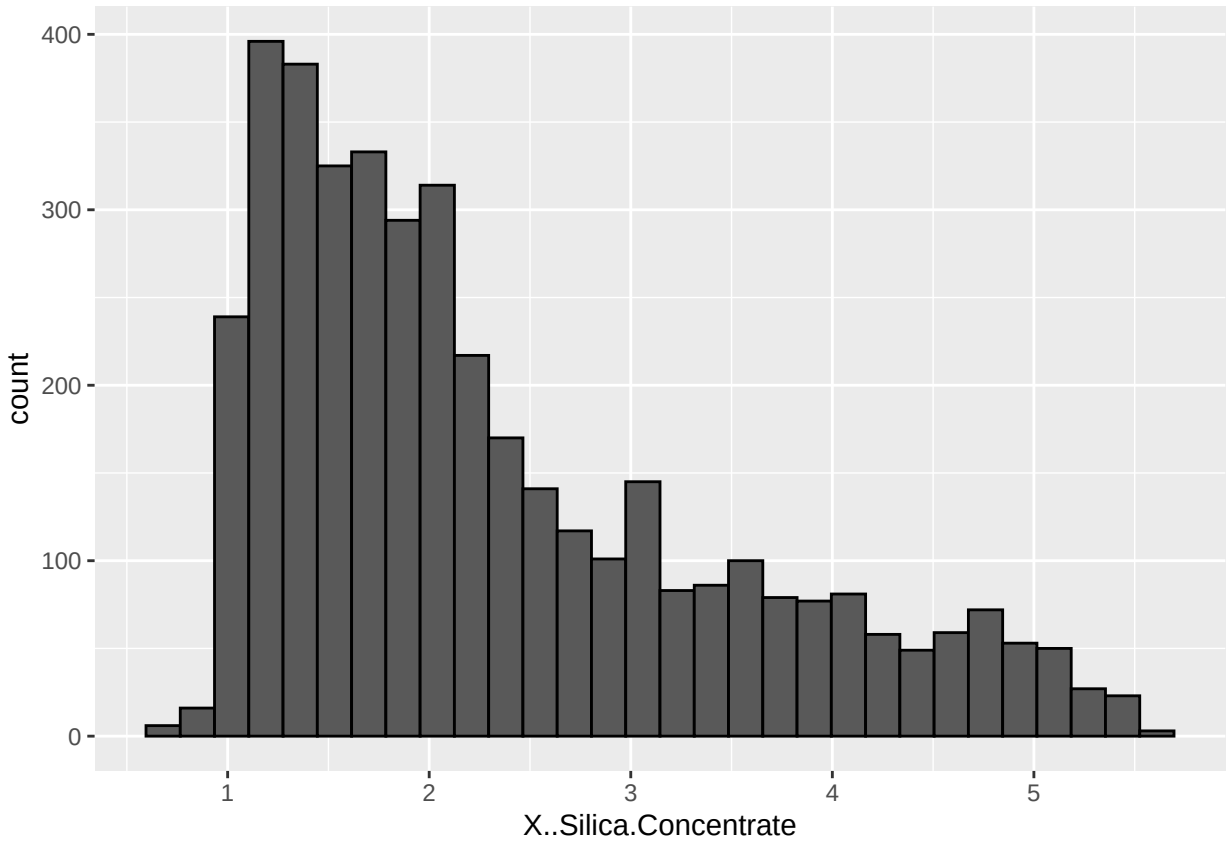
Introduction

The purity of an iron ore concentrate is determined by the amount of silica content contained in it. This is because silica is considered the main contaminant of iron ore. This therefore means that the higher the amount of silica found in the concentrate,the more impure that concentrate and purifying it means getting as little silica in the concentrate, as possible. The amount of silica in an iron ore concentrate is obtained in a process called Flotation- a process in which crushed iron ore bearing rocks are passed through specific conditions in a flotation plant in order to concentrate and extract the iron ore. This project attempts to use those specific conditions as explanatory variables to predict a response variable, silica content, in the concentrate. Being able to predict the amount of silica in the concentrate is necessary as obtaining readings from the laboratory takes a minimum of one hour per sludge fed into the flotation plant. This wastes a lot of time and the ability to accurately predict those readings may help the engineers working in the plant alter conditions so as to improve on ore quality,well in time. The data set used here is from Kaggle and is on this link:link. The data set contains 24 variables,and variables 1 through 23 are explanatory, while variable 24 is the response variable. Readings for all explanatory variables are taken every 20 seconds and those of the response variable are taken every hour. This means that the data set shows the same reading of "X..Silica.Concentrate" for 180 entries, which is the average of the 20s intervals from the explanatory variables.

Method

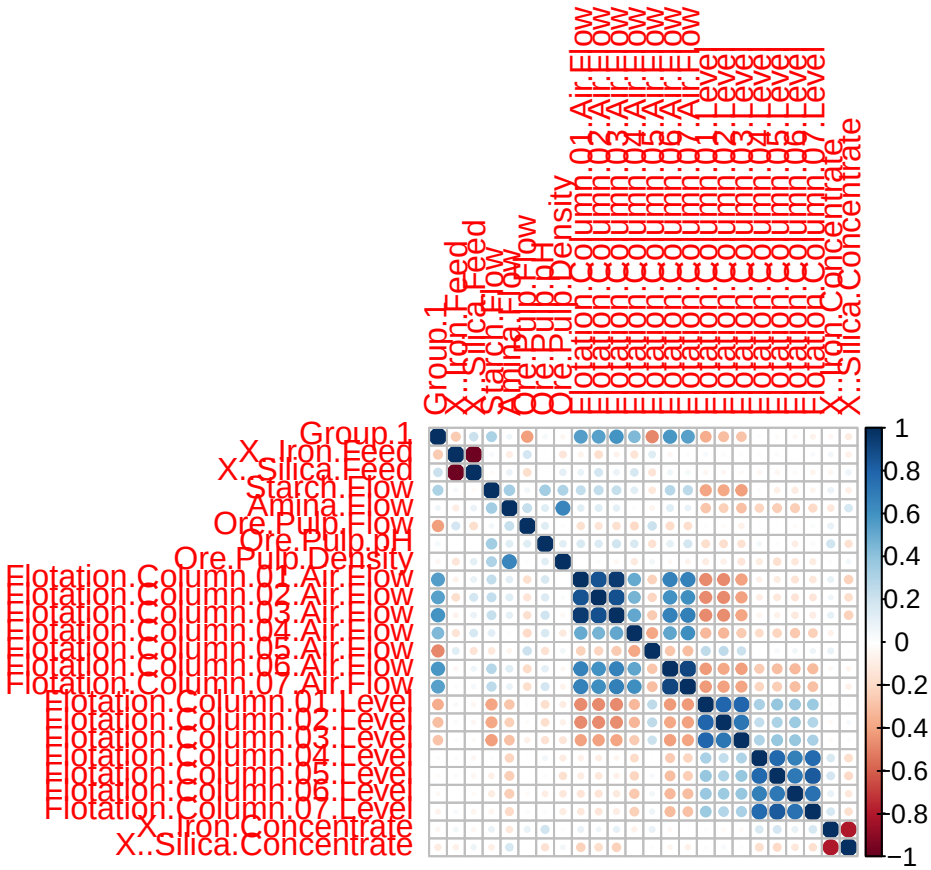
All the variables in the data set are continuous, so I will start with a simple linear regression model to perform the predictions and then categorize the response variable so that I can use other classification algorithms to do the predictions, and then evaluate which perform better.It makes sense to categorize the response variable by creating a cutoff at which I will call the values below "okay",and those above "too large". This is an approach that can be useful practically as the laboratory technicians have a standard, of the value of silica content that they will accept from the plant.

First, I will examine the distribution of the response variable to see whether or not it is changing over time, as that will be an indication that some or combinations of some explanatory variables are probably causing it to change. Below is the distribution of the response variable:



The distribution shows an overall variable silica concentrate content. It also shows a dichotomy; values that may be called "okay", at around 2.5 and below, and those that may be called "too large"; those greater than 2.5, giving confidence in the idea to categorize.

Next I will examine the data set to see whether or not any direct correlations exist between the explanatory variable(s) and the response variable. Below is a correlation plot:



With the exception of the iron content in the concentrate, the correlation plot shows no direct correlations between any of the explanatory variables and the response variables.However, this does not mean that a linear model cannot be used here as the response variable may still be a linear combination of these explanatory variables, so I will go ahead and fit a linear model and see how well it fits.

Linear Regresion Model

The data set was randomly split into training and test sets in order to evaluate the out of sample performance of the models so as to avoid over-training. Also, since a strong correlation, albeit negative, was seen between the iron and silica contents in the concentrate, the iron content in the concentrate was left out of the explanatory variables used in this model. The proposed model then becomes: **Y=m+b_1x_1+b_2x_2+b_3x_3 . . . +b_22x_22 + e**,

where Y is the response variable, b_1 through b_22 are the coefficients to the variables and x_1 through x_22 are the explanatory variables in the data set, m is the intercept and e is the error that the model cannot explain. The correlation plot above showed some correlation between variables, and this was taken into account in building this model, by calling the pca function to first pick up correlations between variables before fitting the model.Also,visual inspection of the data set showed that although not exactly the same, a lot of values in various variables were similar, and so this was also accounted for by calling the zv (zero variance) function to remove those values that were the same before fitting the model. Ten fold cross validation was also used as an attempt to improve how the model fits. The code to do the above tasks is shown in the script. Below is the summary statistics of the fitted linear regression model:

```
##
## Call:
## lm(formula = .outcome ~ ., data = dat, myControl = ..1)
```

```
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.3982 -0.7130 -0.2254  0.5795  3.4210
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  2.331395   0.022394 104.107  < 2e-16 ***
## Group.1                     -0.137535   0.041589  -3.307 0.000959 ***
## X..Iron.Feed                -0.046649   0.107472  -0.434 0.664291
## X..Silica.Feed               0.009787   0.104399   0.094 0.925321
## Starch.Flow                 -0.070940   0.030490  -2.327 0.020081 *
## Amina.Flow                   0.365052   0.033759  10.813  < 2e-16 ***
## Ore.Pulp.Flow               -0.069260   0.028152  -2.460 0.013967 *
## Ore.Pulp.pH                 -0.149818   0.026085  -5.743 1.07e-08 ***
## Ore.Pulp.Density            -0.188228   0.032937  -5.715 1.26e-08 ***
## Flotation.Column.01.Air.Flow -0.417780   0.095546  -4.373 1.29e-05 ***
## Flotation.Column.02.Air.Flow  0.007508   0.050942   0.147 0.882851
## Flotation.Column.03.Air.Flow  0.006990   0.105185   0.066 0.947023
## Flotation.Column.04.Air.Flow  0.100286   0.030011   3.342 0.000848 ***
## Flotation.Column.05.Air.Flow -0.072233   0.027516  -2.625 0.008728 **
## Flotation.Column.06.Air.Flow  0.153558   0.058825   2.610 0.009109 **
## Flotation.Column.07.Air.Flow -0.026046   0.058357  -0.446 0.655416
## Flotation.Column.01.Level     0.005976   0.045891   0.130 0.896410
## Flotation.Column.02.Level    -0.049419   0.040337  -1.225 0.220659
## Flotation.Column.03.Level    -0.028833   0.039982  -0.721 0.470891
## Flotation.Column.04.Level    -0.009601   0.040266  -0.238 0.811557
## Flotation.Column.05.Level    -0.128272   0.047594  -2.695 0.007095 **
## Flotation.Column.06.Level     0.055251   0.036753   1.503 0.132914
## Flotation.Column.07.Level    -0.047329   0.048893  -0.968 0.333158
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.013 on 2025 degrees of freedom
## Multiple R-squared:  0.2027, Adjusted R-squared:  0.1941
## F-statistic:  23.4 on 22 and 2025 DF,  p-value: < 2.2e-16
```

These summary statistics show that although the overall model is statistically significant- the p-value is well below 0.05- and may be used in prediction exercises, not all the variables are significant. Some variables have a coefficient that is not significantly different from 0.

Generalized linear model (glm)

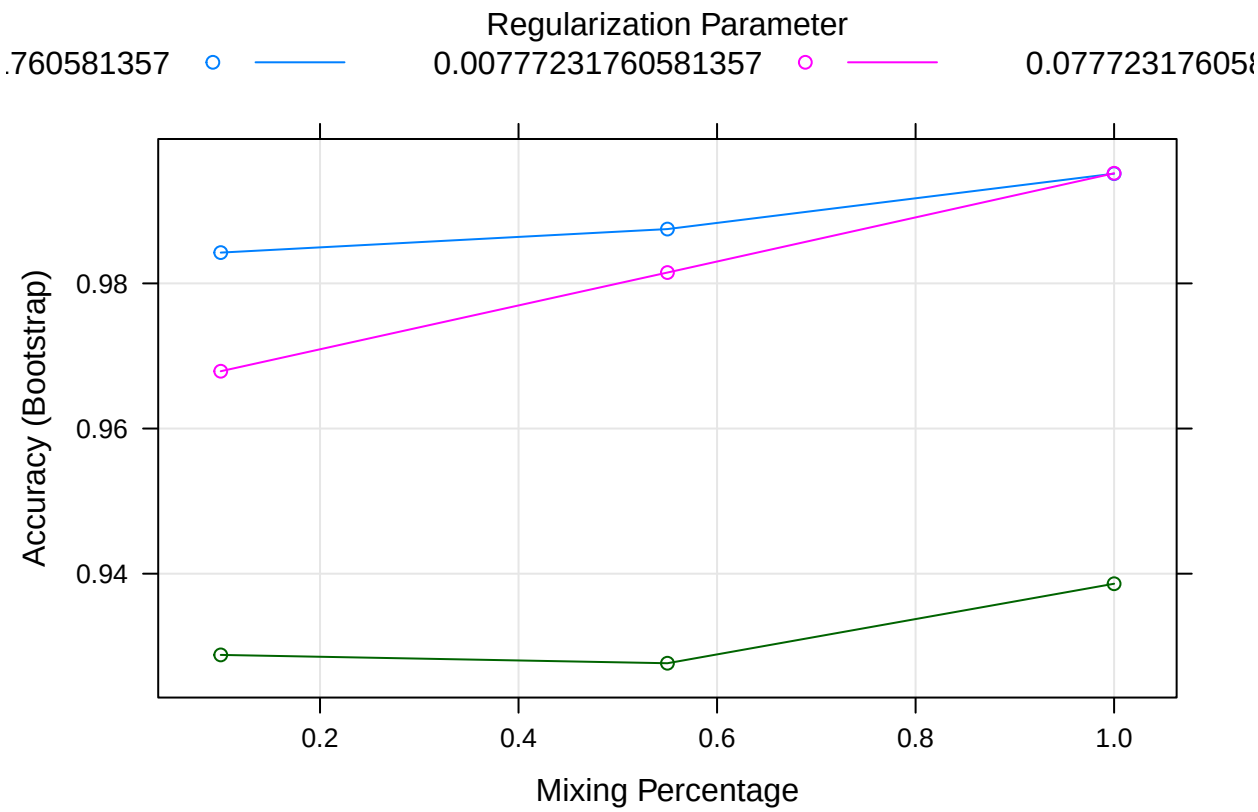
As mentioned earlier in this report, the response variable was categorized into two levels: "okay" and "too_large", which describe the amount of silica content in the ore concentrate. This was done so that classification algorithms such as the generalized linear model may be used. As with the linear model, the iron content in the ore concentrate was dropped as an explanatory variable due to it's high correlation with the response variable. The function pca was also called to preprocess the correlations between other variables before running the algorithm. The code is shown in the script and the results are discussed in the results section below.

K-nearest neighbors model (knn)

Another algorithm that was used is the k-nearest neighbors (knn). The same measure of removing the iron content in the concentrate was taken. This algorithm was used to build a model that will be used to compare with those of the other algorithms used so as to see which performs better. If the knn performs better than the regression algorithms, that would be indicative that the data is not linear. The code is shown in the script and the results in the results section of the report.

Using the glmnet algorithm

The glmnet algorithm is an extension of the glm that penalizes the coefficients of the explanatory variables. It combines the LASSO, which penalizes the number of non-zero coefficients, and RIDGE, which penalizes the absolute magnitude of coefficients, regression methods in one algorithm. Since some of the explanatory variables in the data set are correlated, this algorithm will help deal with that collinearity of those variables as well. This algorithm will also be helpful because the size of the data set is not particularly big. The code for building a model using this algorithm is shown in the script, and the results in the results section of the report. Below is a plot of a fitted glmnet model, it shows how the two regression methods were combined and which one performed better.



The plot shows lasso (pink) and ridge (blue) regressions combined together. The ridge method seems to have performed better, but both methods together improved the accuracy of the model significantly.

Results

Below is the table of results from the various methods used above:

```
##                              Accuracy
## linear regression_cv (rmse)    1.0427
## linear regression_pca (rmse)   1.0541
## glm                            0.9960
## glm_pca                        0.9860
## knn                            0.7838
## glmnet                         0.9971
```

Since the rmse in the linear regression models is in the same units as the response variable, both models make an error a little over 1 percentage point in predicting the percent silica content in the concentrate. The minimum value in that vector is about 0.6 and the maximum is about 5.3. 1 percentage point therefore seems like a very huge error and so those models do not perform well. Additionally, the R^2 value of the linear regression models is very low, indicating that the response variable's variation is not well described by the explanatory variables.

For the classification algorithms, the knn is the worst performing, perhaps because of the linear nature of the data, while the algorithm works well for non-linear data. Calling the function pca overall seems to lower the accuracy of the models. The best performing model is the one built using the glmnet algorithm.

Conclusions and future work

Even though the classification models have high accuracy, only default values were used in building these models. No tuning parameters were used to customize the algorithms as that requires more in depth knowledge of the algorithms. Future work may include customizing the algorithms in order to explore more models. Furthermore, more knowledge of the properties of the plant from which the data set came may be useful in understanding why some explanatory variables are correlated the way they are, perhaps other factors such as geographic or meteorological may provide insight as they may have an effect on some of the variables.